CHAPTER 1: SYSTEMS OF LINEAR EQUATIONS

1.1 Linear Equations

A linear equation in n variables has the form:

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b$$

A system of linear equations is a collection of one or more linear equations involving the same variables.

Example system (2 equations, 2 unknowns):

$$2x + 3y = 8$$
$$x - y = 1$$

Solution Methods:

- Substitution: solve one equation, substitute into other
- Elimination: add/subtract equations to eliminate variables
- Matrix methods: Gaussian elimination, Cramer’s rule

1.2 Gaussian Elimination

Steps:

1. Write augmented matrix [A|b]
2. Use row operations to reach row echelon form:
   - Swap two rows
   - Multiply a row by a nonzero scalar
   - Add a multiple of one row to another
3. Back-substitute to find the solution

Row Echelon Form:

- All zero rows at the bottom
- Leading entry (pivot) of each row is to the right of the pivot in the row above
- All entries below pivots are zero

Solution Types:

- Unique solution: one solution exists
- Infinitely many: free variables exist
- No solution: inconsistent system (0 = nonzero)

CHAPTER 2: MATRICES

2.1 Matrix Basics

A matrix is a rectangular array of numbers arranged in rows and columns. An m x n matrix has m rows and n columns.

```
  A = | 1  2  3 |
      | 4  5  6 |    (2 x 3 matrix)
      | 7  8  9 |
```

Special Matrices:
- Square: m = n
- Identity (I): 1s on diagonal, 0s elsewhere
- Zero matrix: all entries are 0
- Diagonal: nonzero only on main diagonal
- Symmetric: $A = A^T$

2.2 Matrix Operations

Addition: A + B (add corresponding entries)

```
  | 1 2 |   | 5 6 |   | 6  8 |
  | 3 4 | + | 7 8 | = | 10 12 |
```

Scalar Multiplication: k * A (multiply every entry by k)

Matrix Multiplication: AB

  $(AB)_{ij}$ = sum over k of (A_{ik} * B_{kj})

  For A (m x n) and B (n x p), result is (m x p)

  NOTE: AB is NOT generally equal to BA

  Example:

```
  | 1 2 |   | 5 6 |   | 19 22 |
  | 3 4 | * | 7 8 | = | 43 50 |
```

Properties:
- A(B + C) = AB + AC (distributive)
- (AB)C = A(BC)   (associative)
- AI = IA = A   (identity)
- A(BC) = (AB)C   (associative)

2.3 Transpose

The transpose of A (written A^T) is obtained by swapping rows and columns.

```
If A = | 1 2 3 |    then A^T = | 1 4 |
       | 4 5 6 |               | 2 5 |
                               | 3 6 |
```

Properties:

(A^T)^T = A
(A + B)^T = A^T + B^T
(AB)^T = B^T * A^T

CHAPTER 3: DETERMINANTS

3.1 Definition and Computation

The determinant is a scalar value that encodes important properties of a square matrix.

2x2 Matrix:

```
det | a  b | = ad - bc
    | c  d |
```

```
Example: det | 3  2 | = (3)(4) - (2)(1) = 10
             | 1  4 |
```

3x3 Matrix (cofactor expansion along first row):

det(A) = a11*C11 - a12*C12 + a13*C13
where C_ij is the minor (determinant of submatrix)

3.2 Properties of Determinants

- det(A^T) = det(A)
- det(AB) = det(A) * det(B)
- det(kA) = k^n * det(A) for n x n matrix
- det(A) = 0 if and only if A is singular (non-invertible)
- det(I) = 1
- Swapping two rows changes the sign of det
- Adding a multiple of one row to another does not change det

3.3 Invertibility

A square matrix A is invertible (non-singular) if and only if:

- det(A) != 0
- rank(A) = n (full rank)
- The system Ax = 0 has only the trivial solution
- The columns of A are linearly independent

Finding the inverse (2x2):

$$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad A^{-1} = (1/\det) * \begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$$

CHAPTER 4: VECTOR SPACES

4.1 Vectors

A vector is an ordered list of numbers:

$v = (v_1, v_2, ..., v_n)$ in R^n

Vector Operations:

$u + v = (u_1+v_1, u_2+v_2, ..., u_n+v_n)$

$k * v = (k*v_1, k*v_2, ..., k*v_n)$

Dot product: $u \cdot v = u_1*v_1 + u_2*v_2 + ... + u_n*v_n$

Properties of Dot Product:

$u \cdot v = v \cdot u$ (commutative)

$u \cdot (v + w) = u \cdot v + u \cdot w$ (distributive)

$k(u \cdot v) = (ku) \cdot v$

$v \cdot v = |v|^2$ (magnitude squared)

4.2 Linear Independence

Vectors $v_1, v_2, ..., v_k$ are linearly independent if the only solution to $c_1*v_1 + c_2*v_2 + ... + c_k*v_k = 0$ is $c_1 = c_2 = ... = c_k = 0$.

If a nontrivial solution exists, the vectors are linearly dependent.

4.3 Basis and Dimension

A basis for a vector space V is a set of linearly independent vectors that span V (every vector in V can be written as a linear combination of the basis vectors).

The dimension of V is the number of vectors in any basis.

$\dim(R^n) = n$

$\dim(P_n) = n+1$ (polynomials of degree $\leq n$)

Standard basis for R^3: $\{(1,0,0), (0,1,0), (0,0,1)\}$

CHAPTER 5: EIGENVALUES AND EIGENVECTORS

5.1 Definitions

For an n x n matrix A, a scalar lambda is an eigenvalue
and a nonzero vector v is a corresponding eigenvector if:
  Av = lambda * v

Finding Eigenvalues:
  Solve the characteristic equation: det(A - lambda*I) = 0

Example:
```
  A = | 4  1 |
      | 2  3 |

  det(A - lambda*I) = (4-lambda)(3-lambda) - 2*1
    = lambda^2 - 7*lambda + 10 = (lambda-5)(lambda-2) = 0
  Eigenvalues: lambda1 = 5, lambda2 = 2
```

Finding Eigenvectors:
  For each eigenvalue, solve (A - lambda*I)v = 0

```
  For lambda = 5: (A - 5I)v = 0
    |-1  1 | |v1|   |0|
    | 2 -2 | |v2| = |0|
    v1 = v2, so v = t(1,1) for any nonzero t

  For lambda = 2: (A - 2I)v = 0
    | 2  1 | |v1|   |0|
    | 2  1 | |v2| = |0|
    v2 = -2*v1, so v = t(1,-2) for any nonzero t
```

5.2 Diagonalization

A matrix A is diagonalizable if it can be written as:
  A = PDP^(-1)
where D is a diagonal matrix of eigenvalues and P is a
matrix whose columns are the corresponding eigenvectors.

A is diagonalizable if and only if it has n linearly
independent eigenvectors.

Applications:
  - Computing matrix powers: A^k = P * D^k * P^(-1)

- Solving systems of differential equations
- Principal Component Analysis (PCA) in data science